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Intragroup and Intergroup Social Interactions in White-Handed Gibbons

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One of the least studied aspects of the Hylobatidae is their social behavior. In the past, researchers have attributed this to a general lack of affiliative social interactions due to small group size and the hostile nature of intergroup encounters. Data collected on white-handed gibbons (Hylobates lar*) in Khao Yai National Park, Thailand, indicate that gibbons engage in higher rates of social behavior than had previously been appreciated. In addition to relatively high rates of intragroup social contact, gibbons engaged in social behavior, including play and grooming, with members of neighboring groups. These findings obligate a renewed consideration of the nature and function of territoriality in primates.*

KEY WORDS: *Hylobates lar*; play; grooming; social organization; territoriality; intergroup encounters.

INTRODUCTION

Gibbons are small arboreal apes that live in pair-bonded social groups of 2–7 individuals. It is customary to identify the members of the Hylobatidae as being both monogamous and territorial (Chivers, 1977; Leighton, 1987). Recently, however, the value of this blanket characterization has been called into question (Bartlett, 1999b; Fuentes, 1998, 2000; Palombit, 1996; Reichard and Sommer, 1997; Sommer and Reichard, 2000). Palombit (1996) argued that the use of such broad terms to summarize the social system of a whole family of primates conceals more subtle distinctions in the social interactions of the species. For example, Palombit's careful observations of the social

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interactions of white-handed gibbon (*Hylobates lar*) and siamang (*Hylobates syndactylus*) pair mates revealed subtle distinctions between the pair bonds of these closely related species. He concluded that male siamangs, which tend to provide more infant care than white-handed gibbons do, also maintain higher rates of both physical proximity and physical contact with their mates. These distinctions risk being obscured when we categorize all hylobatids as simply monogamous. Others researchers have focused on common usage of the term monogamy as applied to nonhuman primates (Bartlett, 1999b; Fuentes, 1998, 2000; Reichard and Sommer, 1997; Sommer and Reichard, 2000). They concluded that the term is applied inconsistently, being used to describe both a grouping pattern and a mating pattern. In neither case does monogamy characterize the social system exhibited by the hylobatids.

More generally, the recent focus on gibbon social systems highlights a persistent gap in gibbon field research: a lack of studies of gibbon social behavior (Brockelman, 1984). Leighton (1987) described this deficiency as a by-product of group size. The interbirth interval for gibbons is approximately 3 years and sub-adults disperse at *ca.* 9 years of age or older (Brockelman *et al.*, 1998; Brockelman pers. com.). Consequently, it is unusual for a single group to have >3 resident immatures. Compared to animals that live in larger social groups, the availabilty of social partners is limited. An additional consideration is that gibbons tend to defend territorial borders. This has led to the assumption that affiliative social interactions will be limited to direct family members only (Ellefson, 1974; Leighton, 1987). Observtions on white-handed gibbons at Khao Yai National Park in Thailand have shown this assumption to be false. Reichard and Sommer (1997:1150) reported social play between neighboring juvenile gibbons and "close contacts between males and neighboring immatures."

My observations corroborate their findings and supplement them with observations of social grooming between members of neighboring groups. I interpret the details of the interactions in the context of the social history of the population. I also present data that indicate that the overall rate of intra-group social interactions by gibbons may have been underestimated. I argue that the occurrence of affliliative intergroup encounters in gibbons obligates a reevaluation of the nature and evolution of territoriality in primates.

METHODS

Study Site and Subjects

Khao Yai National Park is *ca.* 200 km northeast of Bangkok (101°22′ E, 14°26′ N). Established in 1962, it is the oldest National Park in Thailand and is

the third largest: 2,169 km^2. Most of the park is on a sandstone plateau ranging from 600 to 1000 m above sea level. The Mo Singto study area was first established in 1979 by Warren Brockelman and his students (Brockelman *et al.*, 1998). The site is near the center of the park in an extensive tract of tropical evergreen forest that receives between 2000 and 3000 mm of rainfall annually. Typically the majority of rain falls in 4–5 mo during the southwest monsoon (June through October). There is distinct cool, dry season from November though February, followed by a hot, dry season from March through May (Graham and Round, 1994).

Before the onset of the study 2 white-handed gibbon social groups had been habituated by other observers. They were first described by Raemaekers and Raemaekers (1985) and retain their designations: A and C. A third group, B, whose territory shares a boundary with both groups, was partially habituated. As a result of emigrations, 3 other groups, K, H, and N all had habituated members at the onset of the study (Brockelman *et al.*, 1998).

Data Collection and Analysis

From January 1994 through January 1995 I systematically observed the behavioral ecology of groups A and C. Each group comprised 5 individuals (Table I). The age/sex make-up of each group included 1 adult female, 1 adult male, 1 adolescent (5–8 yrs), 1 juvenile (2–5 yrs), and 1 infant (0–2 yrs) per Brockelman *et al.* (1998). A subadult male emigrated from group C in April 1994.

I followed each of the 2 focal groups from night tree to night tree, whenever possible, for a period of 5 consecutive days per month. During observation days, I recorded the behavior of a single focal subject on the instant

Table I. Composition of gibbon study groups during 1994–95

	Color	Age class	Sex	Name	Comment
Group A	black	adult	male	Fearless	also "F1"
	buff	adult	female	Andromeda	
	black	adolescent	male	Amadeus	
	buff	juvenile	male	Aran	
	buff	infant	female	Akira	b. 10/93
Group C	buff	adult	male	Cassius	
	black	adult	female	Cassandra	
	buff	sub-adult	male	Chet	m. 04/94
	black	adolescent	male	Christopher	
	buff	juvenile	female	Caleba	
	black	infant	female	Cyrana	b. 11/93

b. = born m. = migrated

every 5 min: an instantaneous focal animal sample (Martin and Bateson, 1986). I divided observation days into roughly equal halves and followed a different subject in the morning and afternoon. I rotated data collection to ensure equal representation of all subjects each month, excluding dependent infants. In total, I recorded 14 behavioral categories: and grouped them for analysis as follows: feed (feed and drink), rest (rest and autogroom), travel (travel and move), social (groom, play, and social contact), vocalization, and intergroup encounter. I defined 4 classes of intergroup encounter based on behaviors present during the encounter: agonistic chasing, vocal dispute, neutral, and affiliative.

The results reported here are drawn from 125 observation days, including 109 full-day follows, conducted during the 12 consecutive mo from February 1994 to January 1995. I pooled data for each individual monthly. I used Spearman's correlation coefficient [r_s (Siegel, 1956)] to evaluate the correlation between resource abundance and activity budget (Garber, 1993; Stanford, 1991).

Resource Abundance

I used 2 methods to document seasonal variation in the availability of potential foods. I determined a gross measure of overall resource abundance based on a phenology walk; 252 trees with a DBH $\geq$ 20 cm were marked along 1.5 km of trails. I selected trees without knowledge of species. Once per month I scored all marked trees for the presence of flowers, ripe fruit, and young leaves. I assessed stage of development for fruit and leaves visually based on size and color using 8 $\times$ 30 power binoculars.

During the same interval I surveyed a 3-ha botanical plot for the presence of known gibbon food species. The sample plot represents 6.4% of the combined area of the home ranges of the 2 study groups [46.8 ha (Bartlett, 1999a)]. Each month I walked the borders of 75 20 $\times$ 20-m quadrats noting the presence and location of all fruiting plants, including woody climbers, that accounted for $\geq$5% of the previous month's feeding time.

RESULTS

Resource Abundance

The phenological analysis indicates a bimodal distribution of fruit abundance. The pattern is the same for both the overall abundance of ripe fruit and the abundance of gibbon foods (Fig. 1). The first peak occurred just

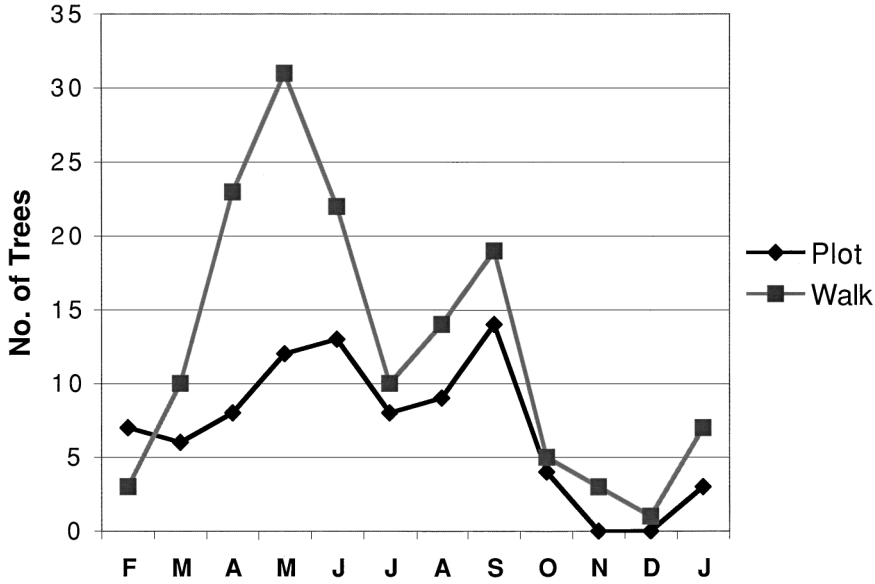


Fig. 1. Comparison of two measures of fruit availability at the Mo Singto study area over a 12-mo period from 1994 to 1995 ($r_s = 0.83$; $p = 0.004$). See text for details.

before the onset of the wet season with a second peak in September. There was a marked reduction in the availability of ripe fruit concomitant with the cool, dry season from November to February. Flowering was strictly seasonal, with peak abundance during the driest months (Table II). Young leaves were relatively abundant throughout the year, though there was a noticeable drop in availability at the beginning of the wet season (June and July).

Intragroup Social Activity

Annual Activity Budget

The majority of the annual activity period was devoted to maintenance behavior (Table III). Feeding, resting and traveling together combined for 82.9% of the annual activity budget. On average, Khao Yai gibbons devoted 11.3% of their activity period to affiliative intragroup social interactions. Time spent in social activity exceeded that spent calling (4.0%) and in intergroup encounters (1.8%). Overall rates of social activity were comparable between adults (10.3%) and immatures, (12.2%; Table IV).

Table II. Overall resource availability based on phenology walk ($N = 252$ Trees)

	No. of trees		
	Fruit	Flowers	Young leaves
Feb '94	3	23	78
Mar	10	17	30
Apr	23	7	64
May	31	8	33
Jun	22	7	13
Jul	10	8	17
Aug	14	6	33
Sep	19	10	37
Oct	5	11	41
Nov	3	31	78
Dec	1	24	46
Jan '95	7	49	62
Mean	11.7	18.9	46.2
Range	1–31	6–49	13–78
S.D.	9.4	14.7	21.9

Social Behavior

Month-to-month variation in social activity was considerable, ranging from a high of 20% during May to a low of 3% in November (Table III). On a seasonal basis, the percentage of the activity budget devoted to intragroup social interactions was greatest during the hot, dry season (March through May) when ripe fruit was abundant: 17.7% of total activity time. Conversely,

Table III. Average percentage (%) of time spent in major activities by Khao Yai gibbons

	Feed	Rest	Travel	Social	Vocal	IGE*
Feb '94	33	16	26	18	4	2
Mar	37	14	29	14	3	3
Apr	27	19	25	19	6	4
May	22	25	25	20	5	3
Jun	26	23	27	16	5	3
Jul	27	25	26	15	5	2
Aug	35	29	23	9	4	0
Sep	36	27	26	6	5	1
Oct	32	35	20	6	4	2
Nov	42	36	18	3	1	0
Dec	38	34	20	5	3	0
Jan '95	35	33	24	5	3	1
Mean	32.5	26.3	24.1	11.3	4.0	1.8
Range	22–42	14–36	18–29	3–20	1–6	0–4
S.D.	5.9	7.4	3.3	6.3	1.3	1.4

*Intergroup Encounter.

Table IV. Average percentage of time spent in major activities by Khao Yai gibbons: Adults vs. immatures

	Adults	Immatures
Feed	31.1	34.0
Rest	31.2	21.1
Travel	20.6	27.7
Social	10.3	12.2
Vocal	4.5	3.5
IGE	2.3	1.4

during the cool try season (November through February), when ripe fruit was scarce, social activity accounted for 7.6% percent of activity time on average. This pattern is consistent between age classes (Fig. 2). On a monthly basis, there is a significant positive correlation between rates of social behavior and the overall availability of ripe fruit ($r_s = 0.63$; $p = 0.036$). When age classes are considered separately the positive correlation does not reach statistical significance (adults, $r_s = 0.55$; $p = 0.070$; immatures, $r_s = 0.49$; $p = 0.107$). Rates of social behavior do not appear to be influenced by the availability of young leaves or flowers.

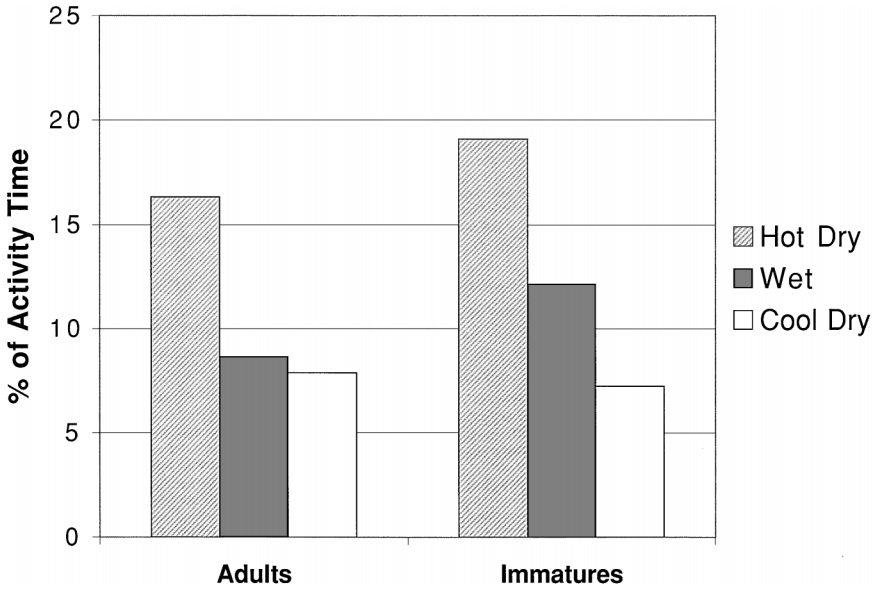


Fig. 2. Comparison of the mean percentage of total activity time devoted to affiliative social behavior by adult and immature gibbons during three seasons.

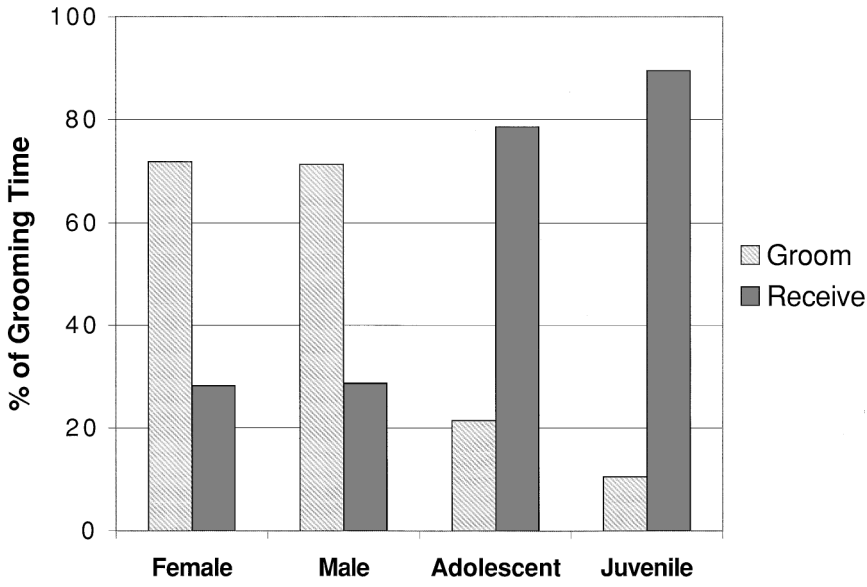


Fig. 3. Mean percentage of focal animal samples spent grooming and being groomed compared across age/sex classes.

Social activity was further divided into 3 classes: groom, play, and social contact. The most common social activity was allogrooming, which represents 63.9% of all social activity, followed by play (29.0%). Social contact (7.1%) was the least common category.

All group members engaged in grooming bouts, which often occurred just after feeding visits and would last for periods $\leq$1 h. Bouts generally consisted of alternating turns, often <1 min in duration, between two individuals. Less frequently $\geq$3 individuals would groom together. This typically occurred when a juvenile joined (or interrupted) an ongoing bout of grooming between two older animals. In dyads involving adults and immatures, the adult spent much more time grooming than being groomed (Fig. 3).

Play behavior, which included wrestling (63.7%), chasing (26.5%), and slapping and biting (9.8%), was primarily exhibited by juveniles and adolescents. Adults of both sexes were occasionally drawn into play bouts by immatures, but more often the adults would groom as the two immatures played together or with the infant. Rates of play were highest among juveniles: 65.9% of all social activity (Fig. 4). On average, immature gibbons (juveniles and adolescents) spent 48.6% of social activity at play; adults spent 9.4%. On an annual basis, rates of play were much more variable than rates of grooming and dropped to near zero from November to January (Fig. 5).

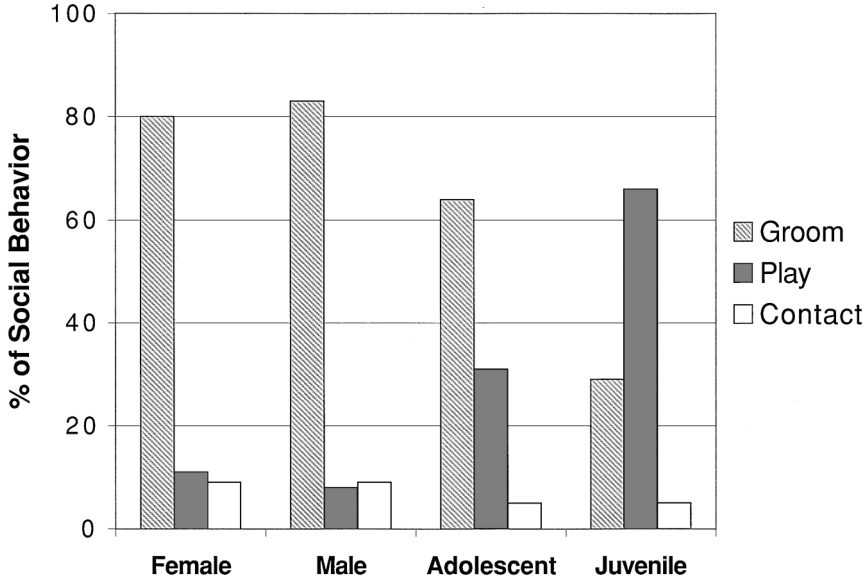


Fig. 4. Comparison of social activity by age/sex class. Values represent the mean percentage of social activity time devoted to the three classes of social behavior.

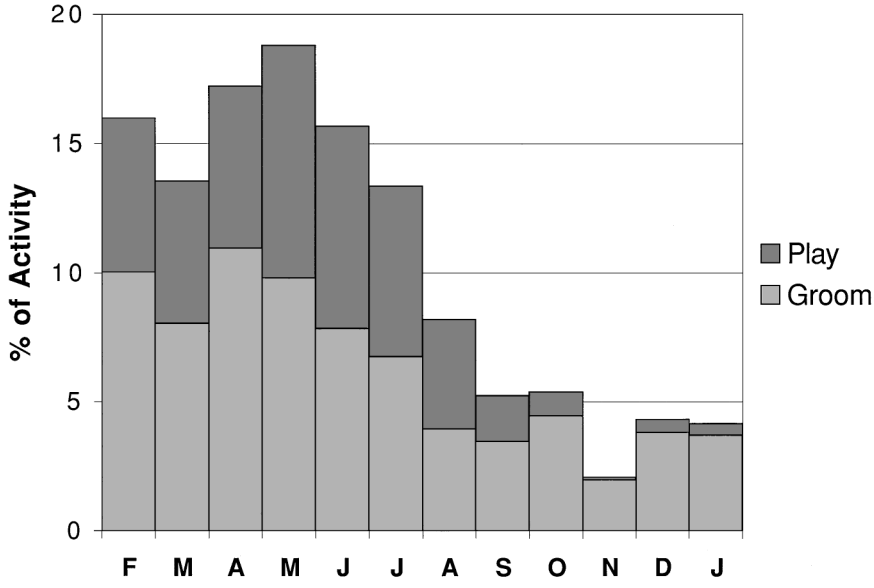


Fig. 5. Monthly variation in rates of play and grooming behavior by Khao Yai gibbons.

Social contact is defined as physical contact between or among individuals while otherwise inactive. Generally it occurred as a prelude to grooming or as an interlude between direction changes during grooming bouts. Less often, during cool wet mornings, two gibbons would sit in contact for several min. On an annual basis social contact accounted for <1% of total activity time.

Intragroup Agonism

Agonistic interactions involving members of the same social group were extremely rare and never recorded during time samples. In total, I recorded 10 instances (over 12 mo) when individuals were aggressively chased, slapped, or grabbed by residents of the same social group. In all instances, the target of the attack was an adolescent male, and all but 2 instances occurred in the context of feeding bouts. In 5 of the 10 cases the aggressor was an adult male, in 3 cases an adult female, and in 1 case an adolescent and a younger juvenile fought over a location in a feeding tree. While they grappled, the resident adult male approached and the adolescent dropped from the crown and did not reenter the tree. Finally, during a single episode, the adolescent of group A was chased after singing a dawn call. In this case the aggressor was not clearly seen.

In addition to overt acts of aggression, subadults and adolescents were occasionally displaced both from feeding sites and from positions in grooming dyads. I have also seen adolescent gibbons remain outside of small feeding trees until other group members had departed.

Encounters Between Social Groups

Intergroup encounters are instances wherein a member or members of a social group achieve(s) sustained visual contact with a member or members of a neighboring group. Encounters are generally preceded by directed movement toward the home range boundary. The average duration of all intergroup encounters was 54 min, with a range of 4 min to 4 hr, 23 min. Encounters between individuals of neighboring groups take several forms. On one end of the continuum members of neighboring groups may engage in social interactions such as play or mutual grooming. On the other end, neighboring males engage in aggressive encounters that may culminate in grappling and biting. Typically, encounters between groups are behaviorally diverse and may include both agonistic and affiliative interactions (Reichard and Sommer, 1997). For the purposes of analysis I identified

mutually exclusive classes of intergroup encounters based on behaviors during the interactions: agonistic, vocal, neutral, and affiliative. Overall, there were 90 group encounters recorded during 109 full-day follows yielding a rate of 0.8 encounters per day. I assigned 87 encounters to 1 of the 4 subclasses; my view of the remaining 3 cases was obstructed.

Agonistic Encounters

The majority of intergroup encounters (58%) were exclusively agonistic and involved chasing between members of neighboring social groups. Much of the time during a dispute is occupied by alternating periods of quiet vigilance or solo calls by males. Stand-offs are punctuated with sudden, vigorous chases that alternate back and forth between the two territories. Agonistic encounters principally involved adult males, though subadults, juveniles, and even adult females occasionally joined disputes.

Vocal Encounters

Twenty percent of intergroup encounters consisted of vocal exchanges only, with one or both males singing extended solos within view of one another. Vocal encounters are distinct from agonistic encounters because they do not involve chasing.

Neutral Encounters

Neutral encounters accounted for 5 instances (6%) of encounters. Neutral encounters represent periods when ≥ 2 neighboring groups fed or foraged in the same trees without engaging in agonistic behavior, sustained vocalizations or affiliative social behavior. All but one of the neutral encounters occurred in the same fruiting fig tree in the area of overlap between gibbon groups A, B, and C.

Affiliative Encounters

All categories of social behavior that occurred between members of the same groups also occurred between members of neighboring groups. In total, 15 of the 87 intergroup encounters (17%) included affiliative interactions

Table V. Affiliative intergroup encounters by gibbon group A and neighboring groups

Group	Date	Interaction	Duration
B	3/21/94	Immatures play chase; A adolescent grooms B juvenile	20 min
B	3/28/94	Immatures play chase, wrestle	52 min
B	5/24/94	Immatures play chase	25 min
C	2/20/94	Immatures play chase	1 hr 20 min
C	6/28/94	Immatures play chase	1 hr 33 min
C	7/23/94	Immatures play chase; C male joins play	44 min
C	7/25/94	Immatures play chase; C male joins play	32 min
C	9/17/94	Juveniles play chase; groups A and C feed	40 min
C	9/26/94	Juveniles play chase	44 min
C	9/28/94	Immatures play chase	16 min
H	4/24/94	Immatures play chase; H male grooms A immature	1 hr 59 min
H	5/24/94	H male grooms A juvenile; H male/A adolescent play chase	1 hr 34 min
H	6/25/94	Juveniles play chase; H male joins play	42 min
H	10/27/94	H male and A juvenile play chase	31 min
H	01/25/95	Immatures play chase	52 min

(Table V). All encounters occurred between group A and one of 3 neighboring groups: B ($N = 3$), C ($N = 7$), and H ($N = 5$). Play, especially chase play between same-aged immatures, occurred during each encounter. Adults generally observed the social interactions of the juveniles and rested in view of the adults of the neighboring group. On 5 occasions, an adult male—either the C male ($N = 2$) or the H male ($N = 3$)—joined the juveniles at play, though adult females never did. An interesting aspect of intergroup play is that it often occurred very low to the ground, especially in and among vines and very small trees. In many cases it appeared that the immatures almost dragged their feet on the ground as they jump and swing between supports.

During 3 encounters gibbons from neighboring groups also engaged in grooming bouts. Once immature gibbons from groups A and B briefly groomed (<5 sec) during a bout of wrestle play. The other cases involved the adult male (Felix) of group H and the immatures of group A. The longest single bout of grooming involved Felix and the adolescent male (Amadeus) of A group. It lasted *ca.* 22 min. At one point Amadeus moved away from Felix into an adjacent tree. Felix followed him, embraced him with both arms and resumed grooming. Afterward they played chase before Amadeus rejoined the A adults in a nearby tree. Upon his return, Amadeus was groomed by the A male. Subsequently, Felix groomed the A juvenile (Aran) for *ca.* 8 min. The encounter ended when the members of group A traveled away from the border area between the two groups. The total duration of the

Table VI. Group encounters by type: Group A and group C[9]

Group	Agonistic	Vocal	Affil	Neutral	Total
Group A					
B	9	2	3	1	15
C	16	4	6	1	27
H			5		5
M	4				4
N	45	2			7
Total	34	8	14	2	58
Group C					
A	7	4	1	2	14
B	8	1		1	10
K	7	4			11
Unknown	3	1			4
Total	25	10	1	3	39

[a] Some encounters involved multiple groups.

encounter was 119 min. The average duration of all affiliative encounters was 50 min.

Although the majority of group encounters were agonistic, those of groups A and H were exceptional; all 5 of their encounters were affiliative (Table VI).

Interactions Between Males and Females During Encounters

While it was uncommon for females to be involved in chases during agonistic intergroup encounters, males showed considerable interest in adult females of neighboring groups. During 17 group encounters, an adult male approached a female of a neighboring group within a few meters, and in 5 encounters the male contacted the female. In 4 cases the C male touched the A female on either her head or hand. For example, during one instance the C male approached the female from above and reached down and brushed his hand through her hair as if grooming. In response, the A female squealed, charged the male and exited the tree. Females were also approached in a more overtly hostile manner. Twice the N male charged the A female during an agonistic encounter, and once the A male pursued the C female, jumped on her back and briefly grappled with her before she turned and chased him away. All male-female encounters occurred while the resident male was out of visual contact with his mate. In every case wherein the resident male returned during an encounter, the rival male was chased from the area.

DISCUSSION

Intragroup Social Behavior

Gibbons have generally been perceived as engaging in very low rates of affiliative social behavior relative to other primates (Ellefson, 1974; Leighton, 1987). An extensive review of gibbon field studies by Leighton (1987:137) concluded:

> [Gibbons] spend surprisingly little time socializing with one another. Singing takes up about 4% of the activity period, grooming and social play usually less. The lack of social partners may contribute to such low levels of interaction.

Studies of three species, in particular, determined that social activity was virtually absent altogether: *H. agilis* (Gittins and Raemaekers, 1980), *H. klossii* (Whitten, 1984), and *H. muelleri* (Leighton, 1987). Long-term studies of lar gibbons have been conducted at three other field sites: Tanjong Triang (Ellefson, 1974) and Kuala Lompat (Gittins and Raemaekers, 1980; Raemaekers, 1979) in Malaysia and Ketambe (Palombit, 1992) in Indonesia. According to Ellefson (1974:82), play accounted for <2% of the activity period of juveniles and grooming accounted for 3% of the activity period of adults. While neither grooming nor play was exclusive to any one age group, he noted that juvenile play bouts showed "some tendency to parallel the adults' grooming bouts." In other words, the level of social activity at Tanjong Triang (2–3%) appears to be consistent with Leighton's conclusion. Similarly, Raemaekers (Gittins and Raemaekers, 1980) reported that lar gibbons at Kuala Lompat devoted 3% of their activity budget to social contact. The value for sympatric siamangs was 2%. Finally, Palombit (1992), who studied lar gibbons at Ketambe did not include social activity in his calculation of the gibbon "activity profile," but characterized the rate of grooming in white-handed gibbons and siamang at Ketambe as "low, but by no means negligible" (Palombit, 1992:302).

Nonetheless, very low rates of social activity are not universal. For example, Chivers (1974:200) found that siamang grooming bouts at Kuala Lompat occupied 15% of activity time. He noted, however, individuals were actually grooming for 42% of the time that they were together: "pair association time." Thus, the value reported by Chivers likely includes time in proximity but not in contact, which would elevate the overall rate of sociality. Morever, Chivers did not identify rates of play behavior. In comparison, the overall rate of social activity, including grooming and play in my sample exceeds 11% of the annual activity budget and the monthly rate ranged as high as 20%. Similarly high rates of social activity have been reported by other observers in Khao Yai (Brockelman *et al.*, 1998; Nettelbeck, 1993; Reichard, 1991; Treesucon, 1984).

There are several possible explanations for the relatively high rates of affiliative social behavior among Khao Yai gibbons. Methodological differences may be a factor. For example, Chivers (1974) and Raemaekers (Gittins and Raemaekers, 1980) come to very different conclusions about the rates of grooming in siamangs. Brockelman *et al.* (1998:337) suggest, "that the interval and scan sampling methods used by most previous observers have tended to underestimate the time spent in play." This conclusion is not consistent with my findings. Alternatively, one substantive way in which my methodology differs is that the data were collected only on focal subjects versus all visible individuals (*cf.* Chivers, 1974; Gittins and Raemaekers, 1980; Whitten, 1984).

Estimates of sociality may also vary depending on when observations are carried out. Rates of social activity vary over the course of the year (Chivers, 1974). At Khao Yai, social interaction was greatest during the hot dry season when ripe fruit was abundant. In April and May, affiliative social behavior accounted for 19% and 20% of the activity period. When fruit was scarce, gibbons exhibited very low rates of social activity (3–5%). One interpretation of these findings is that gibbons minimize non-subsistence activity when high-caloric resources are in short supply in order to conserve energy (Bartlett, 1999a). It appears that the time Khao Yai gibbons devote to play is particularly sensitive to fluctuations in resource abundance. Because play is overwhelmingly a behavior exhibited by immatures, different age classes will be affected differently by fruit scarcity. As a consequence, observations made during different times of the year or on different age classes may yield very different estimates of social activity time. Nevertheless, these factors are unlikely to account for differences between Khao Yai, Kuala Lompat and Tanjong Triang, because in all three cases gibbons were studied for at least a full year and all age groups were sampled.

Differences in the composition and structure of the primate community at different study sites may also affect rates of social activity. For example, Ketambe and Kuala Lompat differ from Khao Yai in that lar gibbons are sympatric with siamang. It is possible that the presence of congeneric competitors increases nutritional stress and thus limits time devoted to social behavior. This hypothesis garners some support from the fact that Khao Yai gibbons are least social during periods of fruit scarcity.

Subjects at different sites may also differ in degree of habituation. Gibbons are notoriously difficult to follow and easy to lose sight of, especially when not traveling or feeding; the sound of fruit fall often indicates their location. Both study groups were well habituated to observers. Group A, in particular, has been the object of thousands of hours of observation over 15 years before this study. In contrast, Ellefson, Palombit and Raemaekers report having to habituate lar gibbons. But it is unlikely that differences in habituation are a major factor. While the study groups at Tanjong Triang

were followed for only 15 mo, Palombit's study continued over 6 years, and Raemaekers reported extremely low rates of social activity for siamang despite his study group (TS1) having been observed by Chivers for over 2 years before his study (Chivers and Raemaekers, 1980).

Differences in reported rates of prosocial behavior may also be contingent on the number of social partners within study groups. Leighton (1987) observed that the rate of social activity is generally lower in groups with smaller numbers of resident immatures. It may be significant that both of the study groups at Khao Yai experienced births just before the onset of data collection. No data were collected on infants, but as they became increasingly independent of the mother, both infants were frequently play partners to older immatures. However, rates of social activity between Khao Yai adults (10.3%) and juveniles (12.2%) did not differ greatly.

A second issue concerns the level of sociality in gibbons relative to other primates. For example, Ellefson (1974:82) concluded, "gibbons play, but they play quantitatively less than many higher primates." When gibbons are compared to the best-studied taxa, e.g. baboons, macaques, and chimpanzees (Strier 1994), this claim may hold true. But, a more productive comparison may be made to atelines, which are similarly arboreal, frugivorous and about the same size as gibbons. Di Fiore and Rodman (2001) reported that *Lagothrix lagotricha* devoted 4.7% of activity time to social behavior, which is <50% of the value for Khao Yai gibbons. More generally, Sussman and Garber (2001) found that, on average, free-ranging primates spend <5% of the time in any type of social interaction (Sussman *et al.*, 2002). Finally, the conclusion that gibbons engage in lower rates of affiliative social behavior relative to other species as a result of smaller group size does not hold across taxa. Kinzey and Wright (1982:269) studied a single family group of 4 titi monkeys (*Callicebus torquatus*) and report that "about 10% of total activity samples during the active day (range 1.9–20.5%) were devoted to giving or receiving grooming." While they do not address rates of play behavior, the pattern of social activity they described is similar to that reported for gibbons: "While the adults groomed each other, the two juveniles were usually separated from them and engaged in bouts of play, interspersed with periods of resting" (p. 269). Kinzey and Wright (1982) also found that the direction of grooming is a function of age. The youngest juvenile received the most grooming and groomed other animals least.

It is evident from the preceding discussion that there is considerable variability across gibbon populations in the amount of the activity period devoted to affiliative social interactions. Nevertheless, based on observations at Khao Yai, it appears that both the absolute and relative level of prosocial behavior in gibbons has been underestimated. Furthermore, while a lack of social partners may contribute to low rates of social behavior relative to

some primate species, there is now substantial data to show that gibbon social interaction, i.e., play and grooming, is not limited to other group members (Jolly, 1972, Table 14; Reichard and Sommer, 1997). Though interactions between adult gibbons are typically hostile, adults show considerable tolerance toward one another during affiliative encounters involving immatures. Habituation of multiple social groups is undoubtedly a prerequisite to successful observations of affiliative interactions between neighboring gibbon groups (Reichard and Sommer, 1997).

Intergroup Encounters

The interactions between Khao Yai groups A and H are of particular interest because all encounters between them were affiliative. I never saw them engage in territorial disputes with one another, which contrasts with groups B and C with which group A had both affiliative and agonistic interactions. To better understand the relationship between groups A and H, it is important to consider their histories (Brockelman *et al.* 1998; Fig. 6).

In 1983, after the death of the resident adult male of group F, the F subadult male (Fearless) displaced the adult male of group A. Subsequently,

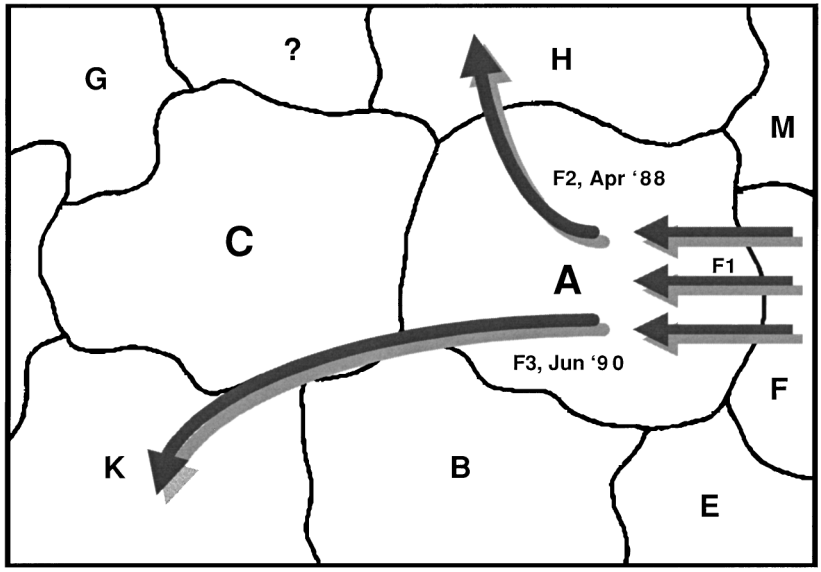


Fig. 6. Summary of group transfer for three subadult males from gibbon group F–Fearless (F1), Felix (F2) and Frodo (F3). All three males immigrated into group A in 1983. The subsequent transfer of two of them into groups H and K is indicated. For details see Brockelman *et al.* (1998).

the adult female of group F emigrated to another group and the 2 immature males of group F followed Fearless and joined group A. By the middle of 1983 group A consisted of the 3 F-group males (Fearless, Felix and Frodo), the A female (Andromeda), and her offspring born in 1980 (Actionbaby). Subsequent offspring were born to the new pair in 1984 (unnamed) and 1987 (Amadeus). Then in 1988, the second oldest F male (Felix), now a subadult, emigrated to the neighboring group H, becoming their resident male. The third, and youngest, F male (Frodo) emigrated into group K in 1990. Finally, additional infants were born to group A in 1990 (Aran) and 1993 (Akira).

It is presumed, though not confirmed, that the two oldest F males, which ultimately became the resident males of neighboring groups A and H, are siblings. In either case, they were both born to group F, and lived together for 5 years in group A. Thus, their familiarity and perhaps genetic relationship may account for the lack of agonistic territorial encounters between the two groups. Moreover, Felix played and groomed with the immatures of group A on 5 occasions. One of them Amadeus, with which he had the longest sustained contact, was born a year before Felix emigrated. In other words, Felix and Amadeus lived together for at least a year and may also be genetically related.

Growing evidence indicates that the complex social history of group A is not unusual. Long-term observations at both Ketambe (Palombit, 1994a) and Khao Yai (Brockelman *et al.*, 1998; Sommer and Reichard, 2000) suggest that many gibbon social groups represent non-nuclear families during a portion of their group history. This has also been confirmed using DNA analysis in the case of at least one group of Bornean gibbons (*Hylobates muelleri*) in Indonesia. Oka and Takenaka (2001:67) report that in one instance, "the family a subadult lived with was not a natal one of the subadult."

Reports of extrapair copulations in both lar gibbons (Reichard, 1995) and siamang (Palombit, 1994b) further strengthen the claim that a nuclear-family model does not accurately characterize gibbon social organization. In light of such findings, several authors have argued that the use of the term monogamy to describe gibbon social systems should either be abandoned outright or modified. Among the alternative labels for gibbon social groups that have been proposed are households (Quiatt, 1985, 1987; Quiatt and Kelso, 1987), two-adult groups (Fuentes, 1998), bisexual clusters (Fuentes, 2000), and grouping monogamy (Sommer and Reichard, 2000).

Regardless of how gibbon groups are labeled, it is becoming increasingly clear that the variability seen in gibbon social systems is, in part, a product of the complex interactions among individuals from neighboring groups. At Khao Yai it is evident that affiliative social interactions between neighbors are to a large extent the result of short dispersal distances by

maturing subadults (Brockelman *et al.*, 1998). The effect is that close kin and prior coresidents occupy neighboring home ranges and thereby maintain long-term affiliative relationships despite the territorial borders that might separate them. These findings punctuate the fact that gibbon groups are not closed social systems per Ellefson (1974:101). Despite being territorial, gibbon family groups are not social isolates (Brockelman, 1984:286; Richard and Sommer, 1997).

If short dispersal distances are the norm for hylobatids (Fuentes, 2000), then it is possible that gibbon social groups (and those of other territorial species) exist as parts of larger stable aggregations that can be identified based on the frequency of affiliative interactions (Rowell, 2000). If so, we may conceive of gibbon populations as being composed of numerous local communities or neighborhoods, which are in turn composed of multiple social groups or households (Quiatt, 1985, 1987). In any case, these findings should refocus our attention on the nature of territoriality in primates. Gibbons show unacknowledged levels of tolerance to neighbors. This fact is inconsistent with models of territoriality as presently conceived (cf. Steenbeek, 1999; van Schaik *et al.*, 1992).

Future research into territoriality in primates should explore the developmental factors that determine tolerance between members of different groups. A major question that remains to be addressed is why some encounters are affiliative while others are hostile. Recalling Richard (1985:335), "The dichotomy between territorial and non-territorial species is incomplete and confusing, and relations between neighboring groups need further study."

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